

Photosynthesis

Photosynthesis is the process by which green plants convert light energy into chemical energy stored in glucose. Photosynthesis takes place mainly in the chloroplasts of leaf cells. The light-dependent reactions of photosynthesis occur in the thylakoid membranes and produce ATP and NADPH. Chlorophyll absorbs mostly red and blue light and reflects green light. Oxygen released during photosynthesis comes from the splitting of water molecules.

Calvin cycle

The Calvin cycle uses ATP and NADPH to fix carbon dioxide into sugar. The enzyme rubisco catalyses the first step of carbon fixation in the Calvin cycle. The Calvin cycle takes place in the stroma of the chloroplast. Each turn of the Calvin cycle fixes one molecule of carbon dioxide. Three turns of the Calvin cycle produce one molecule of glyceraldehyde-3-phosphate.

Cellular respiration

Cellular respiration breaks down glucose to release energy in the form of ATP. Glycolysis is the first stage of cellular respiration and happens in the cytoplasm. The Krebs cycle of cellular respiration occurs in the mitochondrial matrix. The electron transport chain produces most of the ATP during cellular respiration. Oxygen is the final electron acceptor in aerobic cellular respiration.